

Project Design Phase-II
Data Flow Diagram & User Stories

Date	4 May 2026
Team ID	HDI Predictor Team
Project Name	HDI Predictor using Machine Learning and Flask
Maximum Marks	4 Marks

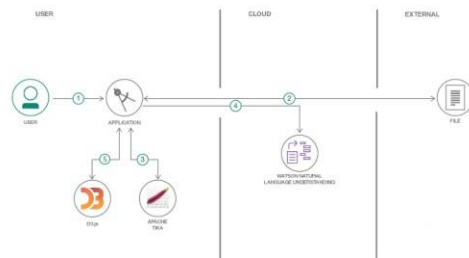
Data Flow Diagrams:

A Data Flow Diagram for the HDI Predictor shows how raw country data moves from the dataset to preprocessing, model training, saved predictions, Flask routes, and the browser interface.

Example: DFD Level 0 (Industry Standard)

Example: (Simplified)

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

User Stories

Use the below template to list all the user stories for the product.

HDI User	Home / Prediction	USN-1	As a user, I can view the HDI Predictor home page and understand the project purpose.	I can open the landing page and read the overview.	High	Sprint-1
HDI User	Home / Prediction	USN-1	As a user, I can view the HDI Predictor home page and understand the project purpose.	I can open the landing page and read the overview.	High	Sprint-1
HDI User	Prediction Input	USN-2	As a user, I can enter life expectancy, schooling, and income values to request an HDI estimate.	The form accepts numeric input and submits successfully.	High	Sprint-1
HDI User	Country Auto-Fill	USN-3	As a user, I can select a country and auto-fill the related indicators from the cleaned dataset.	The selected country populates the form fields.	High	Sprint-1
HDI User	Prediction Output	USN-4	As a user, I can receive an HDI score, category, and confidence value after submission.	The result page shows the prediction summary.	High	Sprint-2
HDI User	Visualization	USN-5	As a user, I can view charts that explain the prediction output.	The charts render correctly on the result page.	Medium	Sprint-2
HDI User	Batch Prediction	USN-6	As a user, I can upload a CSV file and receive predictions for multiple rows.	The CSV endpoint returns a structured prediction table.	High	Sprint-2
Reviewer	Validation	USN-7	As a reviewer, I can test invalid inputs and see friendly error messages.	The application blocks invalid requests.	High	Sprint-3
Reviewer	About Page	USN-8	As a reviewer, I can open the about page and understand the model and technology stack.	The page summarizes the project clearly.	Low	Sprint-3
Developer	Model Loading	USN-9	As a developer, I can load HDI.pkl during app startup.	The model loads without errors.	High	Sprint-3
Developer	Data Flow	USN-10	As a developer, I can trace data from raw CSV to the Flask result page.	The workflow is documented and understandable.	Medium	Sprint-4

HDI User	Home / Prediction	USN-1	As a user, I can view the HDI Predictor home page and understand the project purpose.	I can open the landing page and read the overview.	High	Sprint-1
Developer	Documentation	USN-11	As a developer, I can review the project handouts and report for submission.	The documentation matches the implemented workflow.	Medium	Sprint-4
User	Final Review	USN-12	As a user, I can run the Flask app locally and present the complete HDI Predictor.	The application demonstrates an end-to-end flow.	High	Sprint-4